

Building Progress & Command Logistics

Elevating RTS Immersion through Visual Construction Grids and Recruitment Queues

Executive Overview

Visual Integrity

Replacing static WIP visuals with a dynamic 1x1 tile-based construction system that reflects actual progress on a per-cell basis.

Logistical Depth

Enhancing recruitment via high-fidelity HUDs, avatar integration, and a persistent training queue system with numerical badges.

Construction Grid System

Dynamic Tile Mapping

- Buildings now utilize **House_Construction.png** as a WIP state.
- For N×M buildings, the system maps a grid of 1x1 construction sprites.
- **Idle State:** Grey tint (0x888888) indicates a lack of active labor.
- **Active State:** Natural colors return as workers contribute progress.



Tile-by-Tile Logic

Sequential Completion

Total progress is divided by tile count. If a 2x2 building hits 25%, the first construction tile "finishes" and untints.

Ghost Interaction

Ghost buildings (Placement Mode) show the **final structure** to aid player planning, while placed WIPs show the construction grid.

Hitbox Alignment

Ensures the 128x128 hit area (for 2x2) correctly maps to the four 64x64 construction sub-sprites.

Worker Engagement

Pawn animations sync with individual construction tiles based on distance and task assignment.

Recruitment UX & HUD

Moving from basic buttons to an immersive Selection Panel

Advanced HUD Interface



Command Center Design

- A dedicated bottom-bar panel appears upon selecting a **Barracks**.
- Integrated **Badge Indicators** show queue counts (e.g., "x5").
- Clear visual distinction between Active Training and Queued Units.

Unit **Visual** Identity

Human Avatars

Utilizing the high-quality **Public Assets** avatars for blue faction units:

Warrior_Blue

Archer_Blue

Lancer_Blue

Pawn_Blue



Queue Management

Numerical Stacking

Clicking "Train" increments a queue count. The HUD displays **[Loader] x Number** to reflect the backlog of units.

Cancellation Protocol

Queue items can be removed (popped from the array) by tapping them in the HUD, refunding resources to the player.

```
// State Logic:  
queue: ['warrior', 'warrior', 'warrior'],  
trainingProgress: 0.45 // Progress of the first item
```


Development Steps

Step 1: Refactor BaseBuilding

Implement `renderWIPGrid()` to swap main sprite for tile array during `isCompleted === false`.

Step 2: Training Manager

A global event-driven training loop handles progress and resource consumption per building.

Step 3: React HUD Update

Create `SelectionPanel` component using Tailwind CSS to map `selectedEntity` state to actions.

Step 4: Avatar Store

Load and map avatar textures for display in unit portraits and queue icons.

User Flow Comparison

Current State

Floating "Train" button. No queue visibility. Building looks finished instantly. Floating text loaders.

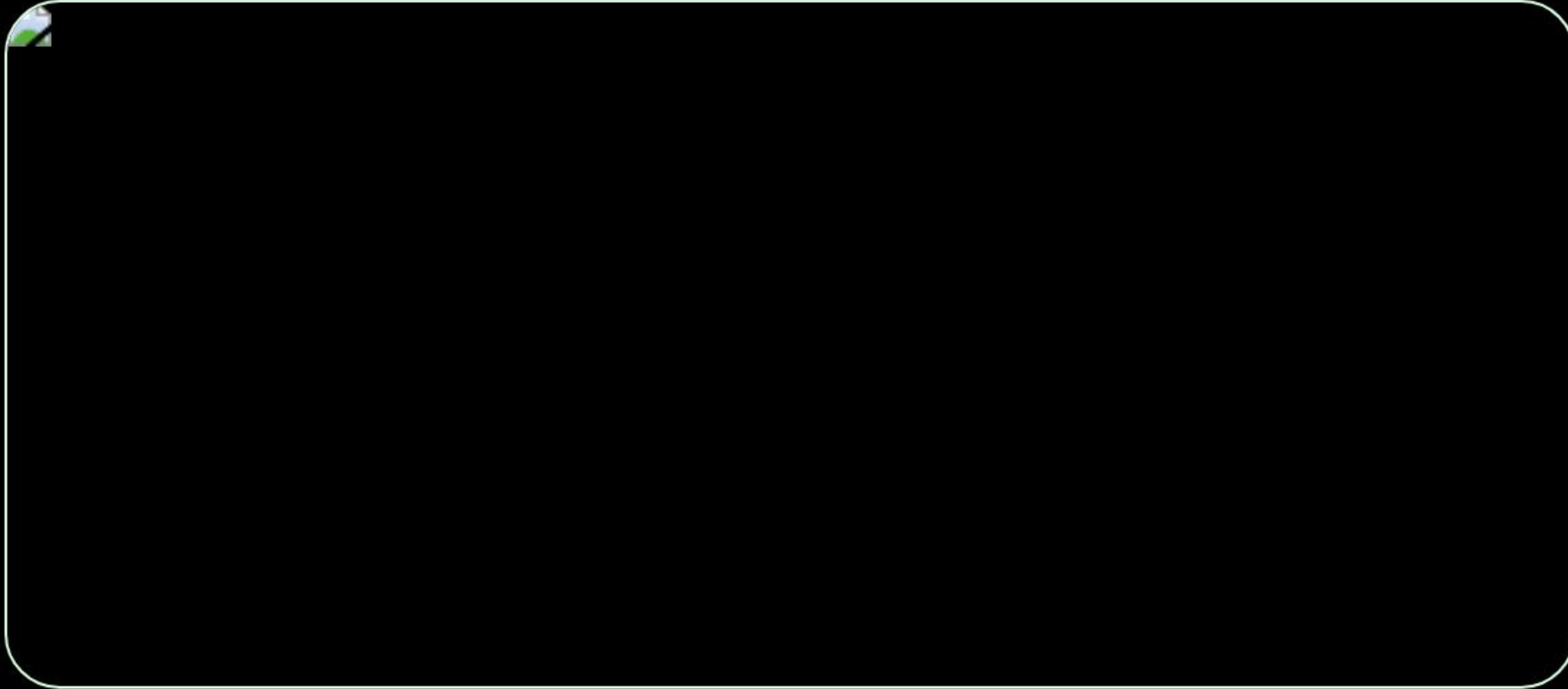
New State

Immersive Bottom Panel. Avatar-based queue with badges. Visual construction grid. Integrated progress rings.

Quick Action Menus

Global Unit Menu

Phase 14 involves a left-screen sidebar for quick-selecting unit types (e.g., selecting all Pawns for emergency gathering).



Lancer, Archer, Monk

Preparation for unique unit classes. Each will require distinct Training Icons and Avatar profiles in the HUD.

Questions & Feedback

Next Steps: Executing the Phase 13.5 Logic Patch

Ready to implement the Construction Grid and Avatar HUD?

Image Sources



https://external-preview.redd.it/construction-system-for-my-medieval-rts-citybuilder-game-v0-yRKwPji3_BAv3ppmodt5sDvUl7pOmOOzbgWaPfnUWOQ.png?format=pjpg&auto=webp&s=7dc6df407e75195889cbad37bd8d22fbb829086a

Source: www.reddit.com



<https://forums.ageofempires.com/uploads/default/original/3X/9/0/9070ba3e3073122e101f7c1a9db89cf8489308d7.jpeg>

Source: forums.ageofempires.com



https://images.stockcake.com/public/e/3/9/e3970c1a-c3ef-4dd0-a9cf-f00945fb879d_large/digital-knight-guardian-stockcake.jpg

Source: stockcake.com



<https://i.ytimg.com/vi/t-D8580Hw18/hq720.jpg?sqp=-oaymwEhCK4FEIIDSFryq4qpAxMIARUAAAAAGAEIAADIQj0AgKJD&rs=AOOn4CLDGqoXblf9A1HlrRP4CJBLdW74eg>

Source: www.youtube.com
